

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **Argentina - Protection of Personal Data (2018)**

Focal Document URL: https://www.argentina.gob.ar/sites/default/files/mensaje_ndeg_147-2018_datos_personales.pdf

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-americas-arg-ppd-2018.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
A	DATA COLLECTION	Processes to ensure completeness and integrity of data, minimise errors, and implement technical measures to ensure confidentiality and limit access during collection.	Functional	Subset Of	Reasonable Data Privacy Practices	PRI-01.11	Mechanisms exist to limit the collection, receiving, processing, storage, transmission, sharing, updating and/or disposal of Personal Data (PD) according to reasonable consumer expectations for what is necessary and proportionate.	10	
A.1	Integrity	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
A.1.1	Ensure Completeness	Verify that the fields in the data collection form allow complete entry of all required data.	Functional	Intersects With	Reasonable Data Privacy Practices	PRI-01.11	Mechanisms exist to limit the collection, receiving, processing, storage, transmission, sharing, updating and/or disposal of Personal Data (PD) according to reasonable consumer expectations for what is necessary and proportionate.	5	
A.1.2	Minimise Entry Errors	Clearly and concisely indicate the type of information to be entered and its format.	Functional	Intersects With	Reasonable Data Privacy Practices	PRI-01.11	Mechanisms exist to limit the collection, receiving, processing, storage, transmission, sharing, updating and/or disposal of Personal Data (PD) according to reasonable consumer expectations for what is necessary and proportionate.	5	
A.1.3	Ensure Integrity	Verify the accuracy of the data entered where the type of record allows (e.g. Date format: DD/MM/YYYY).	Functional	Intersects With	Reasonable Data Privacy Practices	PRI-01.11	Mechanisms exist to limit the collection, receiving, processing, storage, transmission, sharing, updating and/or disposal of Personal Data (PD) according to reasonable consumer expectations for what is necessary and proportionate.	5	
A.1.3	Ensure Integrity	Verify the accuracy of the data entered where the type of record allows (e.g. Date format: DD/MM/YYYY).	Functional	Intersects With	Personal Data (PD) Accuracy & Integrity	PRI-05.2	Mechanisms exist to ensure the accuracy and relevance of Personal Data (PD) throughout the information lifecycle by: (1) Keeping PD up-to-date; and (2) Remediating identified inaccuracies, as necessary.	5	
A.12	Confidentiality	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
A.2.1	Ensure Confidentiality During Collection	Encrypt client-server communication during data collection.	Functional	Intersects With	Transmission Confidentiality	CRY-03	Cryptographic mechanisms exist to protect the confidentiality of data being transmitted.	5	
A.2.2-1	Limit Access to Data Collection	Limit form cache on the client only to the moment of data entry.	Functional	Intersects With	Reasonable Data Privacy Practices	PRI-01.11	Mechanisms exist to limit the collection, receiving, processing, storage, transmission, sharing, updating and/or disposal of Personal Data (PD) according to reasonable consumer expectations for what is necessary and proportionate.	5	
A.2.2-2	Limit Access to Data Collection	Limit data entry on the client to a single user session.	Functional	Intersects With	Reasonable Data Privacy Practices	PRI-01.11	Mechanisms exist to limit the collection, receiving, processing, storage, transmission, sharing, updating and/or disposal of Personal Data (PD) according to reasonable consumer expectations for what is necessary and proportionate.	5	
A.2.3	Limit Unauthorised Access During Collection [DPI]	Use secure digital certificates validated by authorised Certificate Authorities (CA).	Functional	Intersects With	Public Key Infrastructure (PKI)	CRY-08	Mechanisms exist to securely implement an internal Public Key Infrastructure (PKI) infrastructure or obtain PKI services from a reputable PKI service provider.	5	
A.2.3	Limit Unauthorised Access During Collection [DPI]	Use secure digital certificates validated by authorised Certificate Authorities (CA).	Functional	Intersects With	Cryptographic Key Management	CRY-09	Mechanisms exist to facilitate cryptographic key management controls to protect the confidentiality, integrity and availability of keys.	5	
A.2.3-DS	Limit Unauthorised Access During Collection [DSI]	Encrypt communication during transfer from the application server to the database.	Functional	Intersects With	Transmission Confidentiality	CRY-03	Cryptographic mechanisms exist to protect the confidentiality of data being transmitted.	5	
B	ACCESS CONTROL	Implementation of security measures, authentication mechanisms, role and function segregation, and other access characteristics to protect identity and privacy.	Functional	Subset Of	Identity & Access Management (IAM)	IAC-01	Mechanisms exist to facilitate the implementation of identification and access management controls.	10	
B.1	Asset Identification	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
B.1.1	Identify Assets	Prepare an inventory of IT assets that store or manage personal data.	Functional	Intersects With	Asset-Service Dependencies	AST-01	Mechanisms exist to identify and assess the security of Technology Assets, Applications and/or Services (TAAS) that support more than one critical business function.	5	
B.1.1	Identify Assets	Prepare an inventory of IT assets that store or manage personal data.	Functional	Subset Of	Asset Inventories	AST-02	Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that: (1) Accurately reflects the current TAASD in use; (2) Identifies authorized software products, including business justification details; (3) Is at the level of granularity deemed necessary for tracking and reporting; (4) Includes organization-defined information deemed necessary to achieve effective property accountability; and (5) Is available for review and audit by designated organizational personnel.	10	
B.1.1	Identify Assets	Prepare an inventory of IT assets that store or manage personal data.	Functional	Subset Of	Network Diagrams & Data Flow Diagrams (DFDs)	AST-04	Mechanisms exist to maintain network architecture diagrams that: (1) Contain sufficient detail to assess the security of the network's architecture; (2) Reflect the current architecture of the network environment; and (3) Document all sensitive and/or regulated data flows.	10	
B.1.2-1	Define Owners and Responsibilities	Define owners of IT assets that store or manage personal data.	Functional	Subset Of	Asset Ownership Assignment	AST-03	Mechanisms exist to ensure asset ownership responsibilities are assigned, tracked and managed at a team, individual, or responsible organization level to establish a common understanding of requirements for asset protection.	10	
B.1.2-2	Define Owners and Responsibilities	Notify owners of IT assets that store or manage personal data.	Functional	Subset Of	Asset Ownership Assignment	AST-03	Mechanisms exist to ensure asset ownership responsibilities are assigned, tracked and managed at a team, individual, or responsible organization level to establish a common understanding of requirements for asset protection.	10	
B.1.2-3	Define Owners and Responsibilities	Specify access authorisations (type of access and validity) for IT asset owners.	Functional	Intersects With	Role-Based Access Control (RBAC)	IAC-08	Mechanisms exist to enforce Role-Based Access Control (RBAC) for Technology Assets, Applications, Services and/or Data (TAASD) to restrict access to individuals assigned specific roles with legitimate business needs.	8	
B.1.3-1	Verify Application of Controls	Prepare a procedure for periodic inventory updates.	Functional	Intersects With	Updates During Installations / Removals	AST-02.1	Mechanisms exist to update asset inventories as part of component installations, removals and asset upgrades.	5	
B.1.3-1	Verify Application of Controls	Prepare a procedure for periodic inventory updates.	Functional	Intersects With	Standardized Operating Procedures (SOP)	OPS-01.1	Mechanisms exist to identify and document Standardized Operating Procedures (SOP), or similar documentation, to enable the proper execution of day-to-day / assigned tasks.	5	
B.1.3-2	Verify Application of Controls	Prepare an authorisation verification procedure.	Functional	Intersects With	Defining Access Authorizations for Sensitive / Regulated Data	DCH-01.4	Mechanisms exist to explicitly define authorizations for specific individuals and/or roles for logical and /or physical access to sensitive and/or regulated data.	5	
B.1.3-2	Verify Application of Controls	Prepare an authorisation verification procedure.	Functional	Intersects With	Standardized Operating Procedures (SOP)	OPS-01.1	Mechanisms exist to identify and document Standardized Operating Procedures (SOP), or similar documentation, to enable the proper execution of day-to-day / assigned tasks.	5	
B.1.3-3	Verify Application of Controls	Prepare a procedure for new IT assets, defining assigned owner and authorisations.	Functional	Subset Of	Asset Governance	AST-01	Mechanisms exist to facilitate an IT Asset Management (ITAM) program to implement and manage asset management controls.	10	
B.1.3-3	Verify Application of Controls	Prepare a procedure for new IT assets, defining assigned owner and authorisations.	Functional	Intersects With	Standardized Operating Procedures (SOP)	OPS-01.1	Mechanisms exist to identify and document Standardized Operating Procedures (SOP), or similar documentation, to enable the proper execution of day-to-day / assigned tasks.	5	
B.2	Data Access	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
B.2.1-1	Manage System Access	Prepare an internal document defining access controls for each system.	Functional	Intersects With	Applied Security, Compliance and Resilience Controls Documentation	IAO-03	Mechanisms exist to generate authoritative documentation (e.g., System Security Plan (SSP)) that: (1) Identifies key architectural and implementation information on in-scope Technology Assets, Applications and/or Services (TAAS); (2) Reflects the current state of applied security, compliance and resilience controls on applicable People, Processes, Technologies, Data and/or Facilities (PPTDF) that are contained within the system boundary; and (3) Provides a historical record of applied security controls, including changes.	8	
B.2.1-2	Manage System Access	Define and identify superusers (administrators) who can bypass standard access controls.	Functional	Intersects With	Defining Access Authorizations for Sensitive / Regulated Data	DCH-01.4	Mechanisms exist to explicitly define authorizations for specific individuals and/or roles for logical and /or physical access to sensitive and/or regulated data.	5	
B.2.1-2	Manage System Access	Define and identify superusers (administrators) who can bypass standard access controls.	Functional	Intersects With	Privileged Account Management (PAM)	IAC-16	Mechanisms exist to restrict and control privileged access rights for users and Technology Assets, Applications and/or Services (TAAS).	5	
B.2.1-3	Manage System Access	Monitor and track superusers (logging access and activity).	Functional	Intersects With	Privileged Account Management (PAM)	IAC-16	Mechanisms exist to restrict and control privileged access rights for users and Technology Assets, Applications and/or Services (TAAS).	5	
B.2.1-3	Manage System Access	Monitor and track superusers (logging access and activity).	Functional	Intersects With	Privileged User Oversight	MON-01.15	Mechanisms exist to implement enhanced activity monitoring for privileged users.	5	

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B.2.2	Assign Permissions	Provide a concrete and formal notification of responsibilities assumed by each user who internally accesses systems (formal written notification).	Functional	Intersects With	Defined Roles & Responsibilities	HRS-03	Mechanisms exist to define cybersecurity roles & responsibilities for all personnel.	5	
B.2.2	Assign Permissions	Provide a concrete and formal notification of responsibilities assumed by each user who internally accesses systems (formal written notification).	Functional	Intersects With	User Awareness	HRS-03.1	Mechanisms exist to communicate with users about their roles and responsibilities to maintain a safe and secure working environment.	5	
B.2.2	Assign Permissions	Provide a concrete and formal notification of responsibilities assumed by each user who internally accesses systems (formal written notification).	Functional	Intersects With	Formal Indoctrination	HRS-04.2	Mechanisms exist to formally educate authorized users on proper data handling practices for all the relevant types of data to which they have access.	5	
B.2.3-1	Verify Identification and Authorisation	Implement a system that uniquely identifies each user.	Functional	Subset Of	Authenticate, Authorize and Audit (AAA)	IAC-01.2	Mechanisms exist to strictly govern the use of Authenticate, Authorize and Audit (AAA) solutions, both on-premises and those hosted by an External Service Provider (ESP).	10	
B.2.3-2	Verify Identification and Authorisation	Establish a strong password policy.	Functional	Intersects With	Automated System Account Management (Directory Services)	IAC-15.1	Automated mechanisms exist to support the management of system accounts (e.g., directory services).	5	
B.2.3-3	Verify Identification and Authorisation	Maintain an access log for systems.	Functional	Intersects With	System Generated Alerts	MON-01.4	Mechanisms exist to generate, monitor, correlate and respond to alerts from physical, cybersecurity, data protection and supply chain activities to achieve integrated situational awareness.	5	
B.2.3-4	Verify Identification and Authorisation	Maintain a usage log for systems.	Functional	Intersects With	System Generated Alerts	MON-01.4	Mechanisms exist to generate, monitor, correlate and respond to alerts from physical, cybersecurity, data protection and supply chain activities to achieve integrated situational awareness.	5	
B.2.3-5	Verify Identification and Authorisation	Have a procedure for user creation, deletion, and modification.	Functional	Intersects With	Management Approval For New or Changed Accounts	IAC-28.1	Mechanisms exist to ensure management approvals are required for new accounts or changes in permissions to existing accounts.	5	
B.2.3-6	Verify Identification and Authorisation	Limit superuser access to personal data or establish activity monitoring.	Functional	Intersects With	Privileged Account Management (PAM)	IAC-16	Mechanisms exist to restrict and control privileged access rights for users and Technology Assets, Applications and/or Services (TAAS).	5	
B.2.3-7	Verify Identification and Authorisation	Ensure strong password policy is implemented across all systems.	Functional	Intersects With	Password-Based Authentication	IAC-10.1	Mechanisms exist to enforce complexity, length and lifespan considerations to ensure strong criteria for password-based authentication.	5	
B.2.3-8	Verify Identification and Authorisation	Avoid the use of generic/shared user accounts.	Functional	Intersects With	Restrictions on Shared Groups / Accounts	IAC-15.5	Mechanisms exist to authorize the use of shared/group accounts only under certain organization-defined conditions.	5	
B.2.4-1	Control Physical Access to Data Centre	Implement physical access controls at the data centre.	Functional	Intersects With	Physical Access Control	PES-03	Physical access control mechanisms exist to enforce physical access authorizations for all physical access points (including designated entry/exit points) to facilities (excluding those areas within the facility officially designated as publicly accessible).	5	
B.2.4-2	Control Physical Access to Data Centre	Prepare a physical access control procedure.	Functional	Intersects With	Standardized Operating Procedures (SOP)	OPS-01.1	Mechanisms exist to identify and document Standardized Operating Procedures (SOP), or similar documentation, to enable the proper execution of day-to-day / assigned tasks.	5	
B.2.4-2	Control Physical Access to Data Centre	Prepare a physical access control procedure.	Functional	Intersects With	Physical Access Control	PES-03	Physical access control mechanisms exist to enforce physical access authorizations for all physical access points (including designated entry/exit points) to facilities (excluding those areas within the facility officially designated as publicly accessible).	5	
B.2.4-3	Control Physical Access to Data Centre	Maintain a log of physical access (identifying date, time, individuals entering, and reason).	Functional	Intersects With	Controlled Ingress & Egress Points	PES-03.1	Physical access control mechanisms exist to limit and monitor physical access through controlled ingress and egress points.	5	
B.2.4-3	Control Physical Access to Data Centre	Maintain a log of physical access (identifying date, time, individuals entering, and reason).	Functional	Intersects With	Physical Access Logs	PES-03.3	Physical access control mechanisms generate a log entry for each access attempt through controlled ingress and egress points.	5	
B.2.4-4	Control Physical Access to Data Centre	Secure the access control logging system.	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.	5	
B.2.4-4	Control Physical Access to Data Centre	Secure the access control logging system.	Functional	Intersects With	Physical Access Logs	PES-03.3	Physical access control mechanisms generate a log entry for each access attempt through controlled ingress and egress points.	5	
B.2.5	Monitor Activity [DP]	Define a procedure for cleaning inactive accounts with access privileges.	Functional	Intersects With	Disable Inactive Accounts	IAC-15.3	Automated mechanisms exist to disable inactive accounts after an organization-defined time period.	5	
B.2.5	Monitor Activity [DP]	Define a procedure for cleaning inactive accounts with access privileges.	Functional	Intersects With	Standardized Operating Procedures (SOP)	OPS-01.1	Mechanisms exist to identify and document Standardized Operating Procedures (SOP), or similar documentation, to enable the proper execution of day-to-day / assigned tasks.	5	
B.2.5-DS-1	Monitor Activity [DS]	Limit internal system access to a single concurrent session per user account.	Functional	Intersects With	Concurrent Session Control	IAC-23	Mechanisms exist to limit the number of concurrent sessions for each system account.	5	
B.2.5-DS-2	Monitor Activity [DS]	Monitor and control user accounts with special privileges; identify them distinctively.	Functional	Subset Of	Privileged Account Management (PAM)	IAC-16	Mechanisms exist to restrict and control privileged access rights for users and Technology Assets, Applications and/or Services (TAAS).	10	
B.2.5-DS-3	Monitor Activity [DS]	Identify and analyse failed authentication attempts.	Functional	Intersects With	System Generated Alerts	MON-01.4	Mechanisms exist to generate, monitor, correlate and respond to alerts from physical, cybersecurity, data protection and supply chain activities to achieve integrated situational awareness.	5	
B.2.5-DS-3	Monitor Activity [DS]	Identify and analyse failed authentication attempts.	Functional	Intersects With	Security Event Monitoring	MON-01.8	Mechanisms exist to review event logs on an ongoing basis and escalate incidents in accordance with established timelines and procedures.	5	
C	CHANGE CONTROL	Implementation of processes to reliably identify every person making changes in production environments containing personal data, ensuring identification, authentication, and appropriate authorisation.	Functional	Subset Of	Change Management Program	CHG-01	Mechanisms exist to facilitate the implementation of a change management program.	10	
C.1	Change Control	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
C.1.1-1	Ensure Changes [DP]	Verify that changes made to production environments maintain and ensure data integrity.	Functional	Intersects With	Control Functionality Verification	CHG-06	Mechanisms exist to verify the functionality of security, compliance and resilience controls following implemented changes to ensure applicable controls operate as designed.	5	
C.1.1-2	Ensure Changes [DP]	Ensure Data Collection measures (Section A) and Access Control measures (Section B) are applied during change processes.	Functional	Subset Of	Change Management Program	CHG-01	Mechanisms exist to facilitate the implementation of a change management program.	10	
C.1.1-3	Ensure Changes [DP]	Maintain a log of verifications and/or tests performed to ensure integrity, availability, and confidentiality of data.	Functional	Intersects With	Test, Validate & Document Changes	CHG-02.2	Mechanisms exist to appropriately test and document proposed changes in a non-production environment before changes are implemented in a production environment.	5	
C.1.1-DS	Ensure Changes [DS]	Designate a person responsible for managing production environments. Implement a change control procedure for production environments.	Functional	Subset Of	Configuration Change Control	CHG-02	Mechanisms exist to govern the technical configuration change control processes.	10	
D	BACKUP AND RECOVERY	Implementation of backup processes to enable correct recovery following an incident preventing access to originally stored information; defining security, dissemination, training, and capacity-building practices for preventive and corrective security incident tasks.	Functional	Intersects With	Data Backups	BCD-11	Mechanisms exist to create recurring backups of data, software and/or system images, as well as verify the integrity of these backups, to ensure the availability of the data to satisfy Recovery Time Objectives (RTOs) and Recovery Point Objectives (RPOs).	5	
D.1	Backup Copies and Recovery Process	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
D.1.1-1	Ensure a Formal Backup and Recovery Process	Implement a data backup procedure identifying: • Type of information to be backed up • Physical medium to be used • Number of backup copies to be made • Frequency of backup executions • Description of the backup process • Backup retention period • Person responsible for backups	Functional	Intersects With	Data Backups	BCD-11	Mechanisms exist to create recurring backups of data, software and/or system images, as well as verify the integrity of these backups, to ensure the availability of the data to satisfy Recovery Time Objectives (RTOs) and Recovery Point Objectives (RPOs).	5	
D.1.1-2	Ensure a Formal Backup and Recovery Process	Define and verify a recovery testing procedure.	Functional	Intersects With	Testing for Reliability & Integrity	BCD-11.1	Mechanisms exist to routinely test backups that verify the reliability of the backup process, as well as the integrity and availability of the data.	5	
D.1.1-3	Ensure a Formal Backup and Recovery Process	Maintain a recovery test log identifying: • Type of information recovered • Location and date of recovery tests • Results of recovery tests • Person responsible for recovery tests • Personnel involved in recovery tests • Notification to the data controller	Functional	Intersects With	Testing for Reliability & Integrity	BCD-11.1	Mechanisms exist to routinely test backups that verify the reliability of the backup process, as well as the integrity and availability of the data.	5	
D.1.1-4	Ensure a Formal Backup and Recovery Process	Maintain an inventory identifying backup copies, their physical location, and the medium on which they are stored.	Functional	Intersects With	Asset Inventories	AST-02	Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that: (1) Accurately reflects the current TAASD in use; (2) Identifies authorized software products, including business justification details; (3) Is at the level of granularity deemed necessary for tracking and reporting; (4) Includes organization-defined information deemed necessary to achieve effective property accountability; and (5) Is available for review and audit by designated organizational personnel.	5	
D.1.2-1	Ensure Access Control of Media [DP]	Apply Access Control measures (Section B) to backup copies.	Functional	Subset Of	Identity & Access Management (IAM)	IAC-01	Mechanisms exist to facilitate the implementation of identification and access management controls.	10	

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STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **Argentina - Protection of Personal Data (2018)**

Focal Document URL: https://www.argentina.gob.ar/sites/default/files/mensaje_ndeg_147-2018_datos_personales.pdf

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-americas-arg-ppd-2018.pdf>

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D.1.2-2	Ensure Access Control of Media [DP]	Encrypt backup copies using secure tools.	Functional	Equal	Cryptographic Protection	BCD-11.4	Cryptographic mechanisms exist to prevent the unauthorized disclosure and/or modification of backup information.	10	
D.1.2-3	Ensure Access Control of Media [DP]	Secure recovery testing environments using the same security measures as a production environment.	Functional	Intersects With	Technology Assets, Applications and/or Services (TAAS) Recovery & Reconstitution	BCD-12	Mechanisms exist to ensure the secure recovery and reconstitution of Technology Assets, Applications and/or Services (TAAS) to a known state after a disruption, compromise or failure.	5	
D.1.2-4	Ensure Access Control of Media [DP]	Securely delete recovered information after its accuracy has been verified during testing.	Functional	Intersects With	Testing for Reliability & Integrity	BCD-11.1	Mechanisms exist to routinely test backups that verify the reliability of the backup process, as well as the integrity and availability of the data.	5	
D.1.2-4	Ensure Access Control of Media [DP]	Securely delete recovered information after its accuracy has been verified during testing.	Functional	Intersects With	System Media Sanitization	DCH-09	Mechanisms exist to sanitize system media with the strength and integrity commensurate with the classification or sensitivity of the information prior to disposal, release out of organizational control or release for reuse.	5	
D.1.2-DS-1	Ensure Access Control of Media [DS]	Implement fire and flood protection measures at the physical media storage site containing backup copies.	Functional	Intersects With	Water Damage Protection	PES-07.5	Facility security mechanisms exist to protect systems from damage resulting from water leakage by providing master shutoff valves that are accessible, working properly and known to key personnel.	5	
D.1.2-DS-1	Ensure Access Control of Media [DS]	Implement fire and flood protection measures at the physical media storage site containing backup copies.	Functional	Intersects With	Fire Protection	PES-08	Facility security mechanisms exist to utilize and maintain fire suppression and detection devices/systems for the system that are supported by an independent energy source.	5	
D.1.2-DS-2	Ensure Access Control of Media [DS]	Store backup copies at a physical location different from the production system.	Functional	Intersects With	Separate Storage for Critical Information	BCD-11.2	Mechanisms exist to store backup copies of critical software and other security-related information in a separate facility or in a fire-rated container that is not collocated with the system being backed up.	5	
D.1.2-DS-3	Ensure Access Control of Media [DS]	Where backup copies are transported, implement a transit registration and control procedure.	Functional	Intersects With	Media Transportation	DCH-07	Mechanisms exist to protect and control digital and non-digital media during transport outside of controlled areas using appropriate security measures.	5	
D.1.2-DS-3	Ensure Access Control of Media [DS]	Where backup copies are transported, implement a transit registration and control procedure.	Functional	Intersects With	Custodians	DCH-07.1	Mechanisms exist to identify custodians throughout the transport of digital or non-digital media.	5	
D.1.2-DS-4	Ensure Access Control of Media [DS]	Secure recovery testing environments using the same security measures as a production environment.	Functional	Intersects With	Technology Assets, Applications and/or Services (TAAS) Recovery & Reconstitution	BCD-12	Mechanisms exist to ensure the secure recovery and reconstitution of Technology Assets, Applications and/or Services (TAAS) to a known state after a disruption, compromise or failure.	8	
D.1.2-DS-4	Ensure Access Control of Media [DS]	Secure recovery testing environments using the same security measures as a production environment.	Functional	Intersects With	Backup & Restoration Hardware Protection	BCD-13	Mechanisms exist to protect backup and restoration hardware and software.	3	
D.1.2-DS-4	Ensure Access Control of Media [DS]	Secure recovery testing environments using the same security measures as a production environment.	Functional	Intersects With	Isolated Recovery Environment	BCD-14	Mechanisms exist to utilize an isolated, non-production environment to perform data backup and recovery operations through offline, cloud or off-site capabilities.	3	
E	VULNERABILITY MANAGEMENT	Implementation of continuous review processes to identify, analyse, evaluate, and remediate all possible vulnerabilities in computerised systems that process information, applying integrity control, logging, traceability, and verification techniques.	Functional	Subset Of	Vulnerability & Patch Management Program (VPM)	VPM-01	Mechanisms exist to facilitate the implementation and monitoring of vulnerability management controls.	10	
E.1	Vulnerability Management	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
E.1.1-1	Prevent Security Incidents by Design	Consider and analyse possible threats to which computerised systems will be exposed.	Functional	Intersects With	Threat Catalog	THR-09	Mechanisms exist to develop and keep current a catalog of applicable internal and external threats to the organization, both natural and manmade.	5	
E.1.1-1	Prevent Security Incidents by Design	Consider and analyse possible threats to which computerised systems will be exposed.	Functional	Intersects With	Threat Analysis	THR-10	Mechanisms exist to identify, assess, prioritize and document the potential impact(s) and likelihood(s) of applicable internal and external threats.	5	
E.1.1-2	Prevent Security Incidents by Design	Prepare a conceptual map showing the information flow between the various computerised systems.	Functional	Subset Of	Network Diagrams & Data Flow Diagrams (DFDs)	AST-04	Mechanisms exist to maintain network architecture diagrams that: (1) Contain sufficient detail to assess the security of the network's architecture; (2) Reflect the current architecture of the network environment; and (3) Document all sensitive and/or regulated data flows.	10	
E.1.1-3	Prevent Security Incidents by Design	Establish a security document indicating the security measures adopted for information systems.	Functional	Intersects With	Applied Security, Compliance and Resilience Controls Documentation	IAO-03	Mechanisms exist to generate authoritative documentation (e.g., System Security Plan (SSP)) that: (1) Identifies key architectural and implementation information on in-scope Technology Assets, Applications and/or Services (TAAS); (2) Reflects the current state of applied security, compliance and resilience controls on applicable People, Processes, Technologies, Data and/or Facilities (PPTDF) that are contained within the system boundary; and (3) Provides a historical record of applied security controls, including changes.	5	
E.1.2-1	Ensure Adequate Protection [DP]	Establish security controls for applications processing personal data, including: • Role and profile segmentation • Secure authentication • Session management (complying with Access Control – Section B) • Application error message management	Functional	Intersects With	Defined Roles & Responsibilities	HRS-03	Mechanisms exist to define cybersecurity roles & responsibilities for all personnel.	5	
E.1.2-1	Ensure Adequate Protection [DP]	Establish security controls for applications processing personal data, including: • Role and profile segmentation • Secure authentication • Session management (complying with Access Control – Section B) • Application error message management	Functional	Intersects With	Security of Personal Data (PD)	PRI-01.6	Mechanisms exist to ensure Personal Data (PD) is protected by logical and physical security safeguards that are sufficient and appropriately scoped to protect the confidentiality and integrity of the PD.	5	
E.1.2-2	Ensure Adequate Protection [DP]	Implement security rules and controls on servers connected to an external network that store or manage personal data, configuring alerts for possible attacks.	Functional	Subset Of	Configure Technology Assets, Applications and/or Services (TAAS) for High-Risk Areas	CFG-02.5	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) utilized in high-risk areas with more restrictive baseline configurations.	10	
E.1.2-3	Ensure Adequate Protection [DP]	Physically or logically segment the entity's network, separating public from private areas.	Functional	Intersects With	Network Segmentation (macrosegmentation)	NET-06	Mechanisms exist to implement network segmentation within network architectures to isolate Technology Assets, Applications and/or Services (TAAS) from other network resources.	5	
E.1.2-4	Ensure Adequate Protection [DP]	Separate Production, QA, Testing, and Development environments. Implement virus prevention controls on servers that store or manage personal data.	Functional	Intersects With	Separation of Development, Testing and Operational Environments	TDA-08	Mechanisms exist to manage separate development, testing and operational environments to reduce the risks of unauthorized access or changes to the operational environment and to ensure no impact to production Technology Assets, Applications and/or Services (TAAS).	8	
E.1.2-5	Ensure Adequate Protection [DP]	Implement attack prevention controls on workstations that manage personal data.	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.	8	
E.1.2-5	Ensure Adequate Protection [DP]	Implement attack prevention controls on workstations that manage personal data.	Functional	Intersects With	Endpoint Protection Measures	END-02	Mechanisms exist to protect the confidentiality, integrity, availability and safety of endpoint devices.	8	
E.1.2-6	Ensure Adequate Protection [DP]	Implement virus prevention controls on workstations that manage personal data.	Functional	Subset Of	Malicious Code Protection (Anti-Malware)	END-04	Mechanisms exist to utilize anti-malware technologies to detect and eradicate malicious code.	10	
E.1.2-7	Ensure Adequate Protection [DP]	Establish and execute a periodic software/hardware update procedure for all equipment.	Functional	Subset Of	Software & Firmware Patching	VPM-05	Mechanisms exist to conduct software patching for all deployed Technology Assets, Applications and/or Services (TAAS), including firmware.	10	
E.1.2-8	Ensure Adequate Protection [DP]	Designate a person responsible for compliance with security measures. Implement an incident audit system with a logging mechanism that enables tracking of events or actions relating to a possible incident (log system).	Functional	Subset Of	Assigned Security, Compliance & Resilience Responsibilities	GOV-04	Mechanisms exist to assign one or more qualified individuals with the mission and resources to centrally manage, coordinate, develop, implement and maintain an enterprise-wide Security, Compliance & Resilience Program (SCRPP).	10	
E.1.2-9	Ensure Adequate Protection [DP]	Synchronise all servers/equipment with a public time server to ensure correct traceability during audits.	Functional	Subset Of	Clock Synchronization	SEA-20	Mechanisms exist to utilize time-synchronization technology to synchronize all critical system clocks.	10	
E.1.2-10	Ensure Adequate Protection [DP]	Implement a reporting process that allows users to flag security events.	Functional	Subset Of	Incident Handling	IRO-02	Mechanisms exist to cover: (1) Preparation; (2) Automated event detection or manual incident report intake; (3) Analysis; (4) Containment; (5) Eradication; and (6) Recovery.	10	
E.1.2-11	Ensure Adequate Protection [DP]	Implement an incident management system capable of showing registration date, relevant documentation, persons involved, and affected assets.	Functional	Subset Of	Incident Response Operations	IRO-01	Mechanisms exist to implement and govern processes and documentation to facilitate an organization-wide response capability for cybersecurity and data protection-related incidents.	10	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: Argentina - Protection of Personal Data (2018)

Focal Document URL: https://www.argentina.gob.ar/sites/default/files/mensaje_ndeg_147-2018_datos_personales.pdf

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-americas-arg-ppd-2018.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
E.1.2-11	Ensure Adequate Protection [DP]	Implement an incident management system capable of showing registration date, relevant documentation, persons involved, and affected assets.	Functional	Intersects With	Incident Handling	IRO-02	Mechanisms exist to cover: (1) Preparation; (2) Automated event detection or manual incident report intake; (3) Analysis; (4) Containment; (5) Eradication; and (6) Recovery.	8	
E.1.2-DS-1	Ensure Adequate Protection [DS]	Establish security controls for applications processing personal data, including: • Database code injection filters • Application code injection filters	Functional	Subset Of	Configure Technology Assets, Applications and/or Services (TAAS) for High-Risk Areas	CFG-02.5	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) utilized in high-risk areas with more restrictive baseline configurations.	10	
E.1.2-DS-2	Ensure Adequate Protection [DS]	Implement network intrusion detection controls.	Functional	Equal	Network Intrusion Detection / Prevention Systems (NIDS / NIPS)	NET-08	Mechanisms exist to employ Network Intrusion Detection / Prevention Systems (NIDS/NIPS) to detect and/or prevent intrusions into the network.	10	
E.1.2-DS-3	Ensure Adequate Protection [DS]	Implement intrusion and/or information leakage detection controls on workstations with access to personal data processing.	Functional	Intersects With	Data Loss Prevention (DLP)	NET-17	Automated mechanisms exist to implement Data Loss Prevention (DLP) to protect sensitive information as it is stored, transmitted and processed.	5	
E.1.4-DS-1	Guarantee Effective and Lasting Measures	Periodically implement internal audit processes to verify compliance with the foregoing, exporting and retaining reports.	Functional	Intersects With	Control Conformity Monitoring	CPL-03	Mechanisms exist to validate that Technology Assets, Applications, Services and/or Data (TAASD) conform to the organization's security, compliance and/or resilience policies, standards and other applicable requirements.	5	
E.1.4-DS-2	Guarantee Effective and Lasting Measures	Conduct external audits to evaluate the security of internal systems.	Functional	Intersects With	Periodic Audits	CPL-02.2	Mechanisms exist to conduct periodic audits of security, compliance and resilience controls to evaluate conformity with the organization's documented policies, standards and procedures.	5	
F	INFORMATION DESTRUCTION	Implementation of data deletion processes to ensure confidential content is properly destroyed using secure erasure methods, with effective process control.	Functional	Subset Of	Secure Disposal, Destruction or Re-Use of Equipment	AST-09	Mechanisms exist to securely dispose of, destroy or repurpose system components using organization-defined techniques and methods to prevent information being recovered from these components.	10	
F.1	Ensure Information Destruction	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
F.1.1	Establish Destruction Model/Format	Establish a data destruction procedure identifying: • Type of information to be destroyed • Medium containing the information • Person responsible for destruction • Description of the process and destruction method used	Functional	Intersects With	System Media Sanitization Documentation	DCH-09.1	Mechanisms exist to supervise, track, document and verify system media sanitization and disposal actions.	8	
F.1.2	Establish Secure Deletion Mechanisms [DP]	Implement a physical or logical information destruction process ensuring complete deletion of information with no possibility of recovery, meeting three criteria: • Irreversibility • Security • Confidentiality	Functional	Intersects With	System Media Sanitization	DCH-09	Mechanisms exist to sanitize system media with the strength and integrity commensurate with the classification or sensitivity of the information prior to disposal, release out of organizational control or release for reuse.	8	
F.1.2-DS-1	Discard Magnetic Media [DS]	Implement a logical destruction process of continuous overwriting so that original data cannot be recovered, enabling reuse of the magnetic medium.	Functional	Intersects With	Physical Media Disposal	DCH-08	Mechanisms exist to securely dispose of media when it is no longer required, using formal procedures.	5	
F.1.2-DS-2	Discard Magnetic Media [DS]	Where logical destruction is not feasible, implement a physical destruction process using degaussing, disintegration, incineration, pulverisation, shredding, or smelting techniques.	Functional	Intersects With	Secure Disposal, Destruction or Re-Use of Equipment	AST-09	Mechanisms exist to securely dispose of, destroy or repurpose system components using organization-defined techniques and methods to prevent information being recovered from these components.	5	
F.1.3	Designate Destruction Responsible	Designate an authorised person for destruction and document their authorisation.	Functional	Subset Of	Defined Roles & Responsibilities	HRS-03	Mechanisms exist to define cybersecurity roles & responsibilities for all personnel.	10	
F.1.4	Monitor the Process	Maintain an inventory identifying destroyed media.	Functional	Intersects With	System Media Sanitization Documentation	DCH-09.1	Mechanisms exist to supervise, track, document and verify system media sanitization and disposal actions.	5	
G	SECURITY INCIDENTS	Handling of security events and consequent incidents that may affect personal data, including detection, assessment, containment, and response, as well as escalation and remediation of the technical and operational environment.	Functional	Subset Of	Incident Response Operations	IRO-01	Mechanisms exist to implement and govern processes and documentation to facilitate an organization-wide response capability for cybersecurity and data protection-related incidents.	10	
G.1	Notification of Security Incidents	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
G.1.1-1	Establish Responsibilities and Procedures	Prepare a security incident management procedure.	Functional	Intersects With	Incident Handling	IRO-02	Mechanisms exist to cover: (1) Preparation; (2) Automated event detection or manual incident report intake; (3) Analysis; (4) Containment; (5) Eradication; and (6) Recovery.	5	
G.1.1-1	Establish Responsibilities and Procedures	Prepare a security incident management procedure.	Functional	Intersects With	Incident Response Plan (IRP)	IRO-04	Mechanisms exist to maintain and make available a current and viable Incident Response Plan (IRP) to all stakeholders.	5	
G.1.1-2	Establish Responsibilities and Procedures	Designate a person responsible for communications.	Functional	Subset Of	Integrated Security Incident Response Team (ISIRT)	IRO-07	Mechanisms exist to establish an integrated team of cybersecurity, IT and business function representatives that are capable of addressing cybersecurity and data protection incident response operations.	10	
G.1.2	Prepare Incident Report	Prepare a security incident report containing at minimum: • Nature of the breach • Category of personal data affected • Identification of affected users • Measures taken by the controller to mitigate the incident • Measures applied to prevent future incidents	Functional	Subset Of	Incident Stakeholder Reporting	IRO-10	Mechanisms exist to timely-report incidents to applicable: (1) Internal stakeholders; (2) Affected clients & third-parties; and (3) Regulatory authorities.	10	
G.1.3	Send Notification	Send an incident notification with the report attached to: • Postal address: Av. Pte. Gral. Julio A. Roca 710, CABA, C1067ABP • Email: incidente.seguridad@saip.gob.ar	Functional	Intersects With	Cyber Incident Reporting for Sensitive / Regulated Data	IRO-10.2	Mechanisms exist to report sensitive and/or regulated data incidents in a timely manner.	5	
H	DEVELOPMENT ENVIRONMENTS	Definition of information system development environments, whether in-house or third-party.	Functional	Subset Of	Technology Development & Acquisition	TDA-01	Mechanisms exist to facilitate the implementation of tailored development and acquisition strategies, contract tools and procurement methods to meet unique business needs.	10	
H.1	Security in Development Environments	N/A	Functional	No Relationship	N/A	N/A	N/A	0	
H.1.1	Implement Secure Development Policy	Use masking or anonymisation techniques for information in development, testing, and QA environments. NOTE: If requirement H.1.1 is not met and personal data is used in development, testing, or QA environments, ALL recommended measures in Sections A, B, C, D, E, F, and G must be considered and applied.	Functional	Intersects With	Masking Displayed Data	DCH-03.2	Mechanisms exist to apply data masking to sensitive and/or regulated information that is displayed or printed.	5	
H.1.1	Implement Secure Development Policy	Use masking or anonymisation techniques for information in development, testing, and QA environments. NOTE: If requirement H.1.1 is not met and personal data is used in development, testing, or QA environments, ALL recommended measures in Sections A, B, C, D, E, F, and G must be considered and applied.	Functional	Intersects With	De-Identification (Anonymization)	DCH-23	Mechanisms exist to anonymize data by removing Personal Data (PD) from datasets.	5	